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Abstract

The development of sixth-generation (6G) wireless communication technologies is expected to enable highly connected, intelligent, and autonomous industrial environments. Smart factories depend on reliable communication between machines, sensors, controllers, and intelligent systems, making network performance an important factor in industrial operations.

This project presents a data-driven analytical framework for investigating the relationship between 6G network-performance characteristics and manufacturing efficiency in smart-factory environments. The framework combines data preprocessing, exploratory data analysis, network Key Performance Indicator (KPI) analysis, manufacturing-efficiency analysis, correlation analysis, machine-learning-based prediction, model evaluation, and interactive monitoring.

Python-based data-processing and machine-learning techniques were used to examine network and manufacturing indicators. Multiple machine-learning approaches were explored, including Logistic Regression, Decision Tree, and Random Forest methods, followed by model evaluation, validation, feature analysis, and prediction-error analysis. The resulting analytical artifacts include network findings, manufacturing findings, correlation results, model results, predictions, prediction errors, and project-level KPIs.

An interactive Streamlit-based monitoring framework was also developed as part of the project to support efficiency prediction, network-performance monitoring, and alert-oriented analysis. The overall framework demonstrates how network-performance data can be combined with manufacturing indicators to support data-driven analysis of smart-factory operations.

Keywords: 6G, Smart Factory, Industrial IoT, Manufacturing Efficiency, Network Performance, Machine Learning, Random Forest, Predictive Analytics, Industry 4.0

1. Introduction

Smart manufacturing is increasingly dependent on communication networks capable of connecting machines, sensors, controllers, robotic systems, and data-driven intelligence. As manufacturing environments become more automated, communication performance becomes closely associated with the ability of industrial systems to exchange information reliably and in a timely manner.

Sixth-generation wireless communication is being investigated as a future communication foundation for advanced industrial automation. Industrial applications require characteristics such as low communication latency, high reliability, high throughput, massive connectivity, and support for intelligent and time-sensitive applications.

In a smart factory, network performance can influence the availability and quality of information used by manufacturing systems. Network conditions can therefore be studied alongside manufacturing indicators to identify relationships between communication performance and operational efficiency.

This project investigates these relationships through a machine-learning-based analytical framework. Rather than treating network performance and manufacturing performance as independent domains, the proposed approach combines both areas into a common analytical workflow.

2. Related Work

Recent research has examined the potential role of 6G wireless communication in industrial automation and smart manufacturing. Industrial applications require communication systems capable of supporting highly connected environments, real-time monitoring, intelligent control, and data-intensive applications.

Zeydan, Arslan, and Turk discussed 6G wireless communications for industrial automation, including industrial scenarios, requirements, and challenges. Their work highlights the importance of communication capabilities for future industrial systems and provides a relevant background for analysing network characteristics in smart-factory environments.

The present project extends this general research direction from an analytical perspective by studying network-performance indicators together with manufacturing-efficiency indicators and applying machine-learning techniques to the resulting data.

3. Problem Statement

Modern smart factories generate large quantities of network and operational data. However, analysing network performance separately from manufacturing performance can make it difficult to understand how communication characteristics relate to manufacturing outcomes.

The problem addressed in this project is:

How can 6G network-performance indicators and manufacturing-efficiency indicators be analysed together to identify meaningful relationships and support machine-learning-based prediction in smart-factory environments?

4. Objectives

The main objectives of the project are:

- 1\.. To analyse network-performance characteristics relevant to 6G-enabled smart factories.
- 2\.. To analyse manufacturing-efficiency indicators.
- 3\.. To preprocess and prepare the project dataset for analysis and machine learning.
- 4\.. To perform exploratory data analysis on network and manufacturing variables.
- 5\.. To investigate relationships between network performance and manufacturing efficiency.
- 6\.. To develop machine-learning models for manufacturing-efficiency prediction.
- 7\.. To evaluate and compare model behaviour.
- 8\.. To analyse prediction errors and important features.
- 9\.. To develop an interactive monitoring framework for network and manufacturing analysis.
- 10\.. To provide a reproducible analytical workflow for future smart-factory research.

5. Dataset and Methodology

The project follows a structured data-analysis and machine-learning workflow.

5.1 Data Preparation

The initial stage involves examining the available data, identifying relevant variables, preparing the dataset, and handling preprocessing requirements.

The preprocessing stage prepares the data for both statistical analysis and machine-learning experiments.

5.2 Exploratory Data Analysis

Exploratory Data Analysis (EDA) is used to understand the characteristics and distributions of the available variables.

The analysis includes examination of network-related indicators, manufacturing-related indicators, relationships between variables, and patterns relevant to manufacturing efficiency.

5.3 Feature Engineering

Relevant network and manufacturing variables are prepared as features for machine-learning analysis.

Feature engineering allows the project to transform the available operational measurements into a format suitable for model training and evaluation.

5.4 Machine-Learning Pipeline

The machine-learning workflow follows the sequence:

Dataset -> Preprocessing -> Feature Selection -> Train/Test Split -> Model Training -> Prediction -> Evaluation -> Error Analysis

The project explores multiple classification approaches, including:

- * Logistic Regression

- * Decision Tree

- * Random Forest

Model evaluation and validation are used to examine predictive behaviour before selecting the model configuration used in the final analytical workflow.

6. 6G Network KPI Analysis

Network-performance analysis focuses on characteristics relevant to communication quality in smart-factory environments.

The project examines network indicators including:

- * Latency

- * Throughput

- * Network performance

- * Connectivity-related characteristics

- * Reliability-related indicators

These indicators are analysed to understand their relationship with manufacturing-related measurements.

In industrial environments, network characteristics are important because machines, sensors, controllers, and monitoring systems depend on communication for exchanging operational information.

7. Manufacturing Efficiency Analysis

Manufacturing analysis focuses on indicators associated with production performance and operational efficiency.

The project examines manufacturing-related measurements such as:

- * Production performance
- * Production speed
- * Manufacturing efficiency
- * Operational measurements

The objective is to identify patterns in manufacturing performance and investigate whether network characteristics exhibit measurable relationships with these indicators.

8. Network and Manufacturing Relationship Analysis

A central component of the project is the analysis of relationships between network-performance indicators and manufacturin